1. Simplify the expression below into the form $a + bi$. Then, choose the intervals that a and b belong to.

$$\frac{-45 - 33i}{8 - i}$$

   A. $a \in [-5.6, -4.35]$ and $b \in [-6, -4.5]$

   B. $a \in [-5.75, -5.4]$ and $b \in [32.5, 33.5]$

   C. $a \in [-6.2, -5.7]$ and $b \in [-4, -3]$

   D. $a \in [-5.6, -4.35]$ and $b \in [-309.5, -308.5]$

   E. $a \in [-327.55, -326.95]$ and $b \in [-6, -4.5]$

---

2. Simplify the expression below into the form $a + bi$. Then, choose the intervals that a and b belong to.

$$(10 - 4i)(3 + 2i)$$

   A. $a \in [32, 40]$ and $b \in [7, 11]$

   B. $a \in [19, 23]$ and $b \in [25, 39]$

   C. $a \in [19, 23]$ and $b \in [-33, -29]$

   D. $a \in [32, 40]$ and $b \in [-11, -4]$

   E. $a \in [25, 35]$ and $b \in [-11, -4]$

---

3. Simplify the expression below and choose the interval the simplification is contained within.

$$17 - 12 \div 16 * 13 - (15 * 10)$$

   A. $[-142.75, -135.75]$

   B. $[-135.06, -131.06]$

   C. $[165.94, 170.94]$

   D. $[-79.5, -71.5]$

E. None of the above

---

4. Choose the **smallest** set of Real numbers that the number below belongs to.

$$-\sqrt{\frac{64}{289}}$$

A. Whole

B. Integer

C. Rational

D. Irrational

E. Not a Real Number

---

5. Choose the **smallest** set of Complex numbers that the number below belongs to.

$$\frac{-13}{0} + \sqrt{187}i$$

A. Rational

B. Irrational

C. Nonreal Compelx

D. Pure Imaginary

E. Not a Complex Number

---

6. Solve the equation below. Then, choose the interval that contains the solution.

$$-19(-11x - 10) = -13(6x + 12)$$

A. $x \in [-1.32, -1.16]$

B. $x \in [-0.05, 0.13]$

C. $x \in [-0.25, -0.05]$

D. $x \in [-0.36, -0.25]$

E. There are no Real solutions.

---

7. Solve the linear equation below. Then, choose the interval that contains the solution.

$$\frac{8x-8}{7} - \frac{3x+8}{4} = \frac{-4x-3}{3}$$

A. $x \in [0.75, 1.3]$

B. $x \in [-1.8, -0.95]$

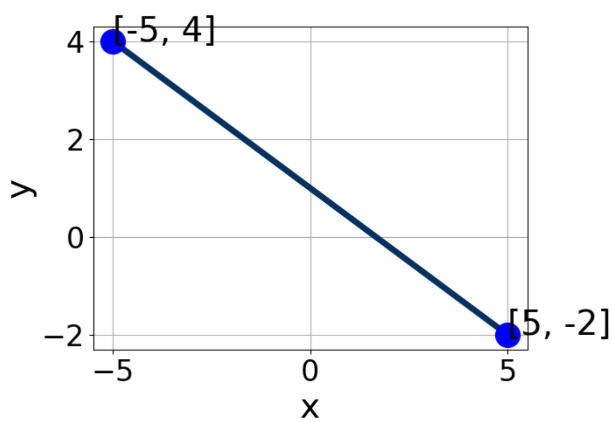
C. $x \in [7.35, 7.8]$

D. $x \in [-0.15, 0.7]$

E. There are no Real solutions.

---

8. Write the equation of the line in the graph below in Standard Form $Ax + By = C$. Then, choose the intervals that contain $A, B,$ and C.



A. $A \in [2.5, 4.5], \quad B \in [3, 6], \text{ and } \quad C \in [4.5, 6.5]$

B. $A \in [-4, -2.5], \quad B \in [-6, -4.5], \text{ and } \quad C \in [-5.5, -4.5]$

C. $A \in [2.5, 4.5], \quad B \in [-6, -4.5], \text{ and } \quad C \in [-5.5, -4.5]$

D. $A \in [0, 2], \quad B \in [0.5, 2], \text{ and } \quad C \in [-0.5, 1.5]$

E. $A \in [0, 2], \quad B \in [-1.5, 0], \text{ and } \quad C \in [-2, 0]$

9. First, find the equation of the line containing the two points below. Then, write the equation in the form $y = mx+b$ and choose the intervals that contain m and b.

$$(-3, 10) \text{ and } (-8, -7)$$

A. $a \in [2.5, 5]$ and $b \in [19.5, 21]$

B. $a \in [-4, -3]$ and $b \in [-35, -33]$

C. $a \in [2.5, 5]$ and $b \in [-22, -19.5]$

D. $a \in [2.5, 5]$ and $b \in [11.5, 13.5]$

E. $a \in [2.5, 5]$ and $b \in [0.5, 2.5]$

10. Find the equation of the line described below. Write the linear equation in the form $y = mx + b$ and choose the intervals that contain m and b.

Perpendicular to $9x - 5y = 7$ and passing through the point $(7, -5)$.

A. $a \in [-1.5, 0]$ and $b \in [-2, -1]$

B. $a \in [-0.5, 2]$ and $b \in [-9.5, -8.5]$

C. $a \in [-1.5, 0]$ and $b \in [-0.5, 2]$

D. $a \in [-1.5, 0]$ and $b \in [-13, -11]$

E. $a \in [-2.5, -1]$ and $b \in [-2, -1]$

11. Using an interval or intervals, choose the option that describes all the x-values within or including a distance of the given values.

Less than 6 units from the number 4.

A. $[-2, 10]$

B. $(-2, 10)$

C. $(-\infty, -2) \cup (10, \infty)$

D. $(-\infty, -2] \cup [10, \infty)$

E. None of the above

---

12. Solve the linear inequality below. Then, choose the constant and interval combination that describes the solution set.

$$-4 - 3x \leq \frac{-12x - 3}{5} < 4 - 3x$$

A. $(a, b]$, where $a \in [3.75, 10.5]$ and $b \in [-9, -6.75]$

B. $[a, b)$, where $a \in [-1.5, 10.5]$ and $b \in [-12.75, -5.25]$

C. $(-\infty, a) \cup [b, \infty)$, where $a \in [3, 7.5]$ and $b \in [-11.25, -6]$

D. $(-\infty, a] \cup (b, \infty)$, where $a \in [0.75, 8.25]$ and $b \in [-12, -3]$

E. None of the above.

---

13. Solve the linear inequality below. Then, choose the constant and interval combination that describes the solution set.

$$-5 + 4x > 5x \text{ or } -3 + 9x < 10x$$

A. $(-\infty, a)$ or (b, ∞), where $a \in [-6, -3.5]$ and $b \in [-4, -2.5]$

B. $(-\infty, a)$ or (b, ∞), where $a \in [1.5, 3.5]$ and $b \in [-5.5, -4]$

C. $(-\infty, a]$ or $[b, \infty)$, where $a \in [-5.5, -3.5]$ and $b \in [-3.5, -2.5]$

D. $(-\infty, a]$ or $[b, \infty)$, where $a \in [2, 4]$ and $b \in [-6.5, -4.5]$

E. $(-\infty, \infty)$

---

14. Solve the linear inequality below. Then, choose the constant and interval combination that describes the solution set.

$$-6x + 9 \leq 10x + 3$$

A. $[a, \infty)$, where $a \in [0.2, 1.3]$

B. $(-\infty, a]$, where $a \in [-0.2, 0.61]$

C. $[a, \infty)$, where $a \in [-2, 0.2]$

D. $(-\infty, a]$, where $a \in [-0.84, -0.16]$

E. None of the above

---

15. Solve the linear inequality below. Then, choose the constant and interval combination that describes the solution set.

$$\frac{3}{3} - \frac{3}{6}x < \frac{6}{5}x - \frac{6}{7}$$

A. (a, ∞), where $a \in [-0.3, 5.2]$

B. $(-\infty, a)$, where $a \in [-0.91, 5.09]$

C. (a, ∞), where $a \in [-3.3, -0.5]$

D. $(-\infty, a)$, where $a \in [-3.09, -0.09]$

E. None of the above

---

16. Solve the quadratic equation below. Then, choose the intervals that the solutions x_1 and x_2 belong to, with $x_1 \leq x_2$.

$$15x^2 + 38x + 24$$

A. x_1 in $[-1.6, 1.01]$ and x_2 in $[-1.22, -1.16]$

B. x_1 in $[-20.64, -19.3]$ and x_2 in $[-18.02, -17.97]$

C. x_1 in $[-6.85, -4.48]$ and x_2 in $[-0.29, -0.05]$

D. x_1 in $[-3.24, -1.62]$ and x_2 in $[-0.75, -0.6]$

E. x_1 in $[-5.12, -3.68]$ and x_2 in $[-0.55, -0.32]$

---

17. Factor the quadratic below. Then, choose the intervals that contain the constants in the form $(ax + b)(cx + d); b \leq d$.

$$24x^2 + 2x - 35$$

A. $a \in [5.32, 8.33], b \in [-17, -6], c \in [3.45, 4.01],$ and $d \in [-5, 7]$

B. $a \in [0.45, 1.82], b \in [-31, -23], c \in [0.9, 1.34],$ and $d \in [27, 32]$

C. $a \in [2.37, 3.15], b \in [-17, -6], c \in [7.84, 8.28],$ and $d \in [-5, 7]$

D. $a \in [11.39, 12.09], b \in [-17, -6], c \in [1.49, 2.25],$ and $d \in [-5, 7]$

E. None of the above.

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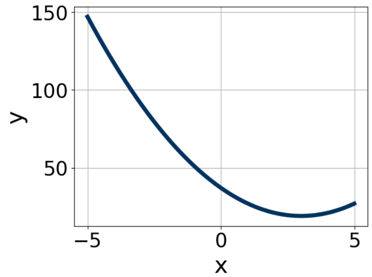
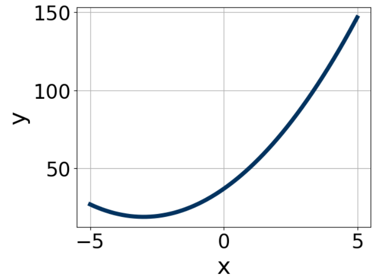
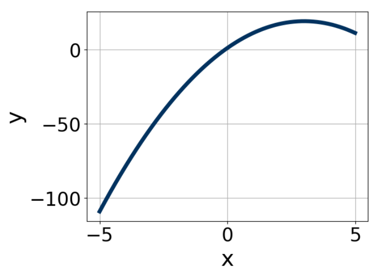
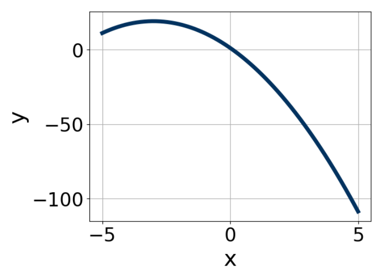
18. Choose the graph of the equation below.

$$f(x) = (x - 3)^2 + 19$$

A. 

C. 

B. 

D. 

E. None of the above.

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19. Solve the quadratic equation below. Then, choose the intervals that the solutions belong to, with $x_1 \leq x_2$ (if they exist).

$$-13x^2 - 7x + 2$$

A. x_1 in $[-0.87, -0.45]$ and x_2 in $[-0.94, 0.58]$

B. x_1 in $[0.22, 0.78]$ and x_2 in $[5.96, 6.97]$

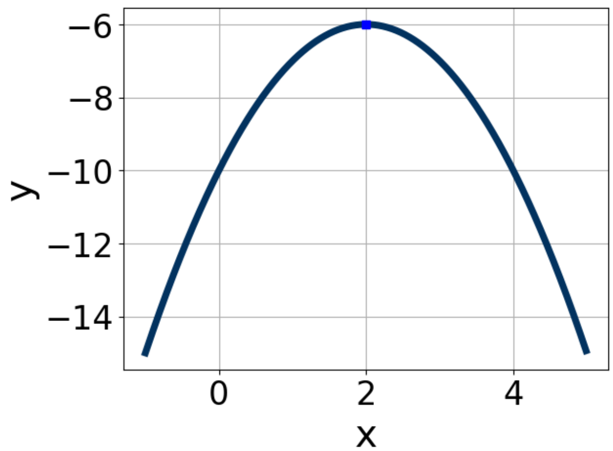
C. x_1 in $[-0.5, -0.09]$ and x_2 in $[0.61, 0.79]$

D. x_1 in $[-12.77, -12.62]$ and x_2 in $[11.45, 12.42]$

E. There are no Real solutions.

---

20. Write the equation of the graph presented below in the form $f(x) = ax^2 + bx + c$, assuming $a = 1$ or $a = -1$. Then, choose the intervals that a, b, and c belong to.



A. $a \in [-1.7, -0.2], \quad b \in [2, 6], \text{ and } \quad c \in [-11, -8]$

B. $a \in [-1.7, -0.2], \quad b \in [-9, -3], \text{ and } \quad c \in [-11, -8]$

C. $a \in [0.1, 1.2], \quad b \in [2, 6], \text{ and } \quad c \in [-3, -1]$

D. $a \in [-1.7, -0.2], \quad b \in [-9, -3], \text{ and } \quad c \in [1, 3]$

E. $a \in [0.1, 1.2], \quad b \in [-9, -3], \text{ and } \quad c \in [-3, -1]$
